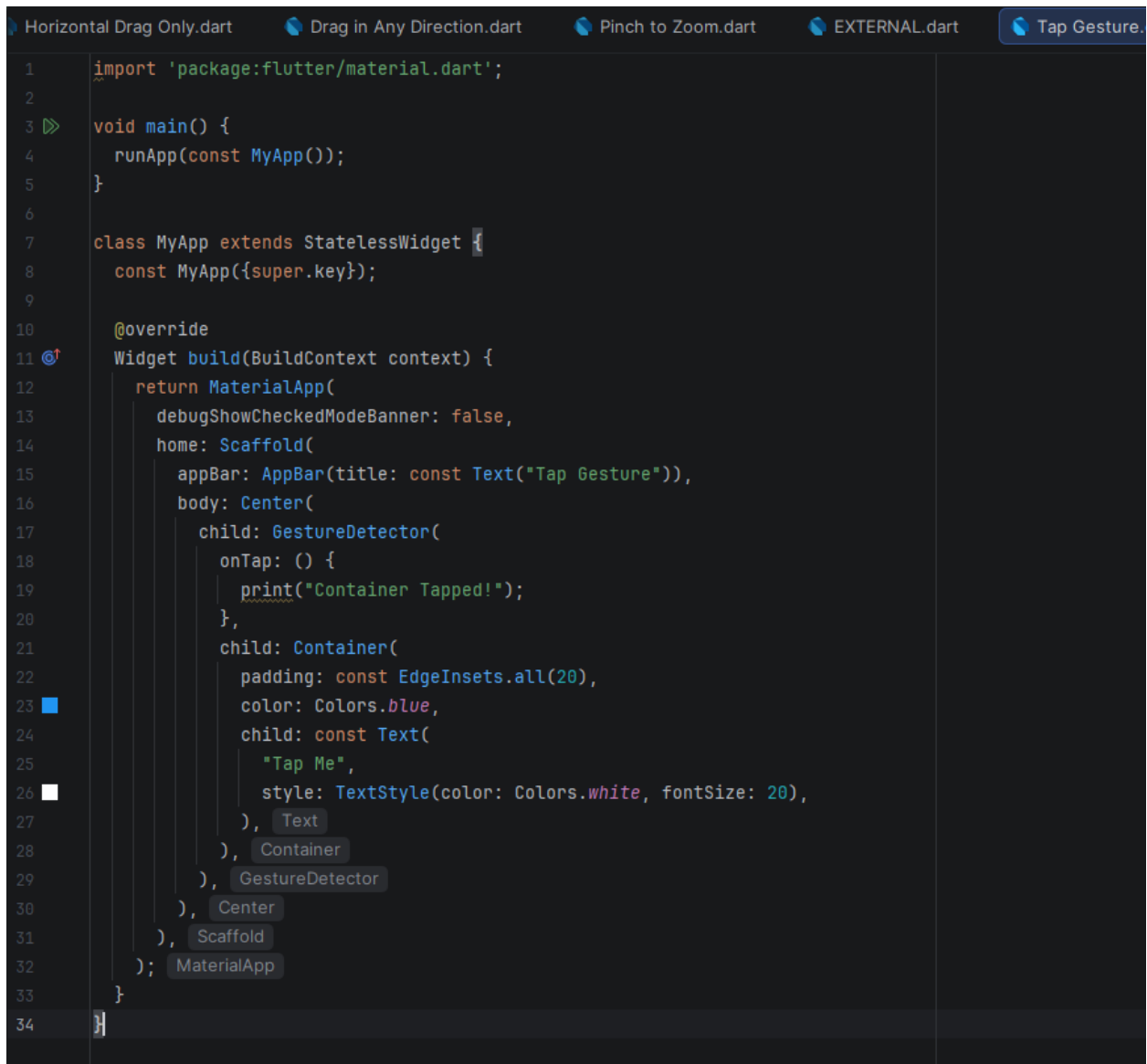


APP DEVELOPMENT WITH FLUTTER

IMPLEMENT DIFFERENT TYPES OF GESTURES IN FLUTTER

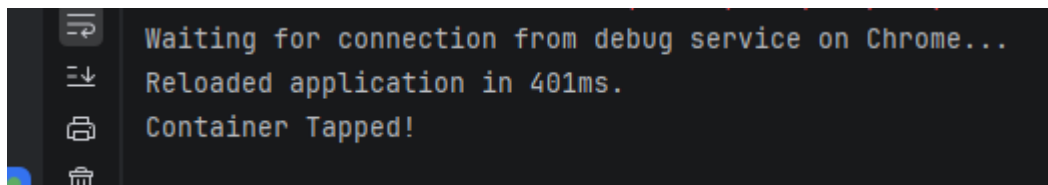
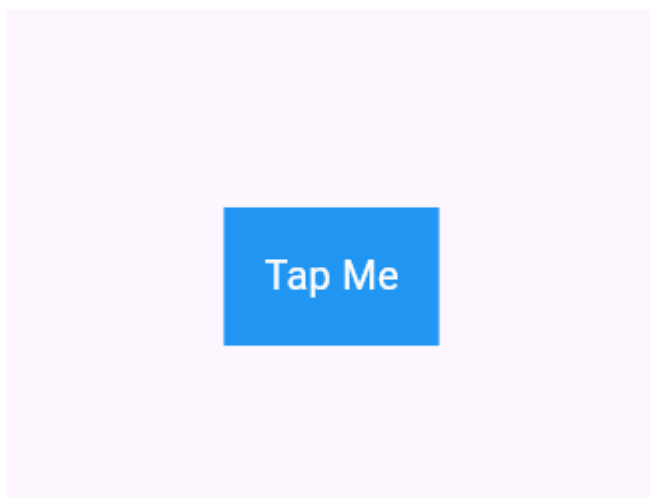
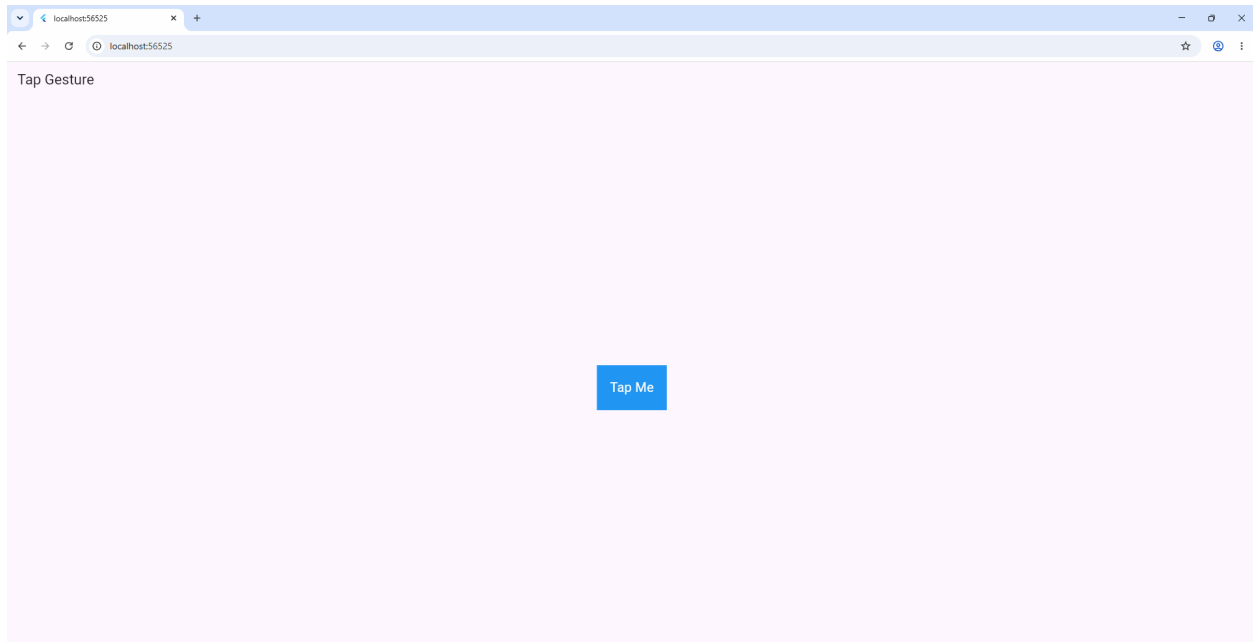
1) SINGLE TAP GESTURE:

CODE:



```
Horizontal Drag Only.dart Drag in Any Direction.dart Pinch to Zoom.dart EXTERNAL.dart Tap Gesture.
1 import 'package:flutter/material.dart';
2
3 void main() {
4   runApp(const MyApp());
5 }
6
7 class MyApp extends StatelessWidget {
8   const MyApp({super.key});
9
10  @override
11  Widget build(BuildContext context) {
12    return MaterialApp(
13      debugShowCheckedModeBanner: false,
14      home: Scaffold(
15        appBar: AppBar(title: const Text("Tap Gesture")),
16        body: Center(
17          child: GestureDetector(
18            onTap: () {
19              print("Container Tapped!");
20            },
21            child: Container(
22              padding: const EdgeInsets.all(20),
23              color: Colors.blue,
24              child: const Text(
25                "Tap Me",
26                style: TextStyle(color: Colors.white, fontSize: 20),
27              ),
28            ),
29          ),
30        ),
31      ),
32    );
33  }
34 }
```

OUTPUT:



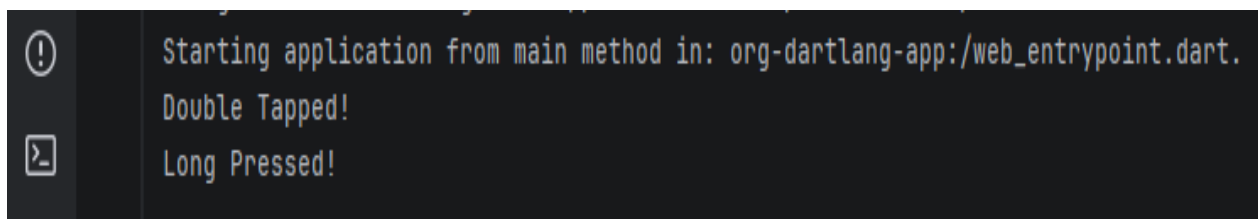
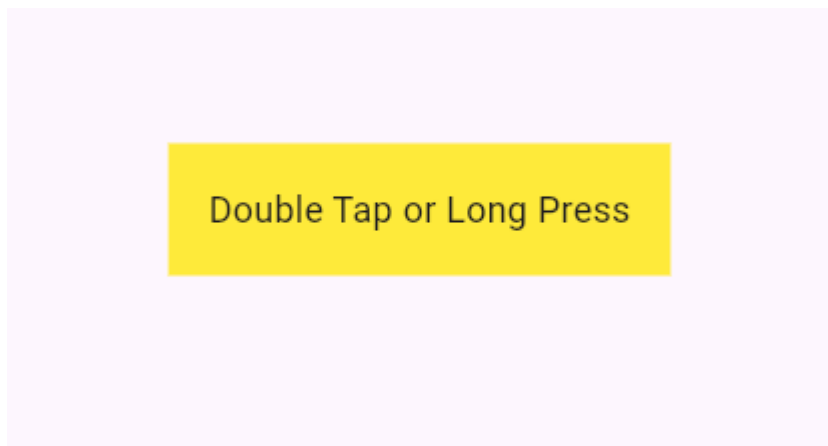
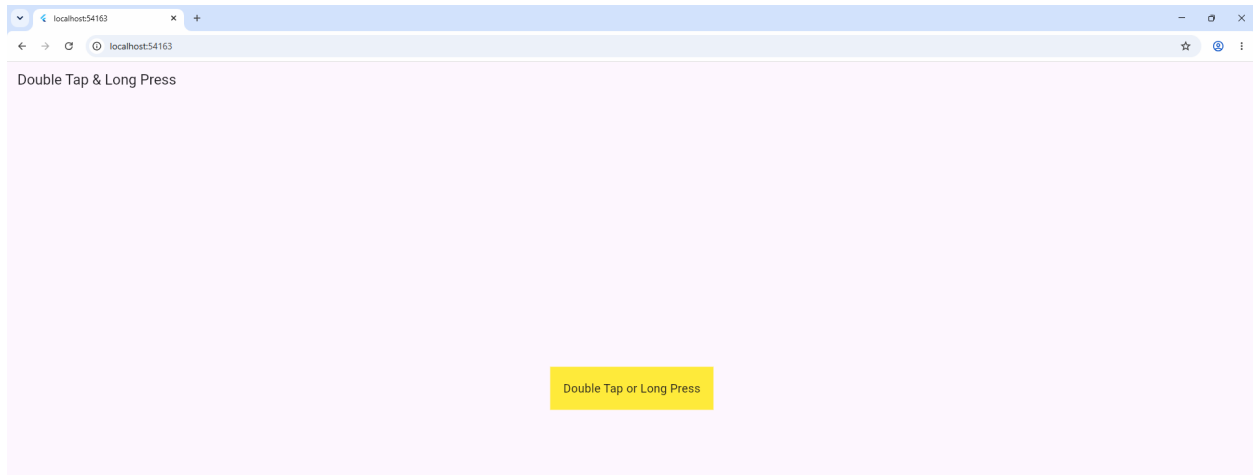
2) DOUBLE AND LONG PRESS GESTURE:

CODE:

```
Double Tap & Long Press.dart × Horizontal Drag Only.dart Drag in Any Direction.dart Pinch to Zoom.dart

1 import 'package:flutter/material.dart';
2
3 void main() {
4   runApp(const MyApp());
5 }
6
7 class MyApp extends StatelessWidget {
8   const MyApp({super.key});
9
10  @override
11  Widget build(BuildContext context) {
12    return MaterialApp(
13      debugShowCheckedModeBanner: false,
14      home: Scaffold(
15        appBar: AppBar(title: const Text("Double Tap & Long Press")),
16        body: Center(
17          child: GestureDetector(
18            onTap: () {
19              print("Double Tapped!");
20            },
21            onLongPress: () {
22              print("Long Pressed!");
23            },
24            child: Container(
25              padding: const EdgeInsets.all(20),
26              color: Colors.yellow,
27              child: const Text(
28                "Double Tap or Long Press",
29                style: TextStyle(fontSize: 18),
30              ),
31            ),
32          ),
33        ),
34      ),
35    );
36  }
37 }
```

OUTPUT:



3) PINCH TO ZOOM GESTURE:

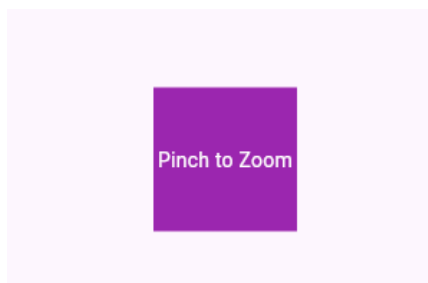
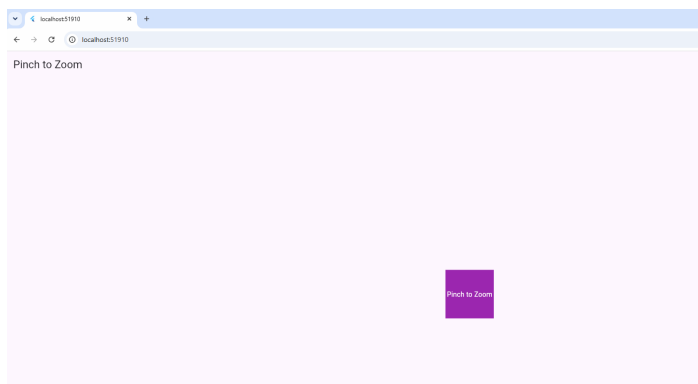
CODE:

```
Double Tap & Long Press.dart Horizontal Drag Only.dart Drag in Any Direction.dart Pinch to Zoom.dart x EXTER...

1 import 'package:flutter/material.dart';
2
3 void main() {
4   runApp(const MyApp());
5 }
6
7 class MyApp extends StatelessWidget {
8   const MyApp({super.key});
9
10  @override
11  Widget build(BuildContext context) {
12    return const MaterialApp(
13      debugShowCheckedModeBanner: false,
14      home: PinchZoomDemo(),
15    );
16  }
17 }
18
19 class PinchZoomDemo extends StatefulWidget {
20   const PinchZoomDemo({super.key});
21
22   @override
23   State<PinchZoomDemo> createState() => _PinchZoomDemoState();
24 }
25
26 class _PinchZoomDemoState extends State<PinchZoomDemo> {
27   double scale = 1.0;
28
29   @override
30   Widget build(BuildContext context) {
31     return Scaffold(
32       appBar: AppBar(title: const Text("Pinch to Zoom")),
33       body: Center(
34         child: GestureDetector(
35           onScaleUpdate: (details) {
36             setState(() {
37               scale = details.scale;
38             });
39           },
40         ),
41       ),
42     );
43   }
44 }
```

```
39     },  
40     child: Transform.scale(  
41       scale: scale,  
42       child: Container(  
43         width: 100,  
44         height: 100,  
45         color: Colors.purple,  
46         alignment: Alignment.center,  
47         child: const Text(  
48           "Pinch to Zoom",  
49           style: TextStyle(color: Colors.white),  
50           textAlign: TextAlign.center,  
51         ), Text  
52       ), Container  
53     ), Transform.scale  
54   ), GestureDetector  
55 ), Center  
56 ); Scaffold  
57 }  
58 }
```

OUTPUT:



4) HORIZONTAL DRAG ONLY:

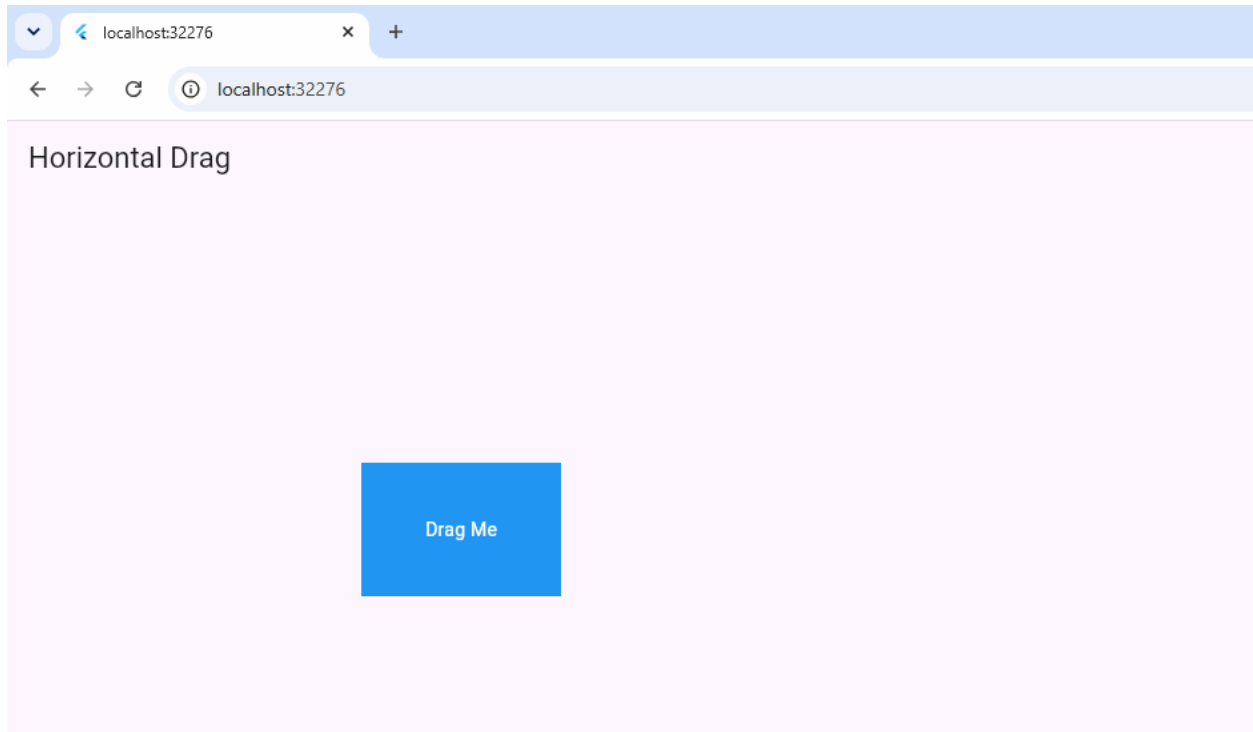
CODE:

```
Double Tap & Long Press.dart Horizontal Drag Only.dart x Drag in Any Direction.dart Pinch to Zoom.dart EXTE

1 import 'package:flutter/material.dart';
2
3 void main() {
4   runApp(const MyApp());
5 }
6
7 class MyApp extends StatelessWidget {
8   const MyApp({super.key});
9
10  @override
11  Widget build(BuildContext context) {
12    return const MaterialApp(
13      debugShowCheckedModeBanner: false,
14      home: DragDemo(),
15    );
16  }
17 }
18
19 class DragDemo extends StatefulWidget {
20   const DragDemo({super.key});
21
22   @override
23   State<DragDemo> createState() => _DragDemoState();
24 }
25
26 class _DragDemoState extends State<DragDemo> {
27   double left = 0;
28 }
```

```
28
29 @override
30 Widget build(BuildContext context) {
31   return Scaffold(
32     appBar: AppBar(title: const Text("Horizontal Drag")),
33     body: Stack(
34       children: [
35         Positioned(
36           left: left,
37           top: 200,
38           child: GestureDetector(
39             onHorizontalDragUpdate: (details) {
40               setState(() {
41                 left += details.delta.dx;
42               });
43             },
44             child: Container(
45               width: 150,
46               height: 100,
47               color: Colors.blue,
48               alignment: Alignment.center,
49               child: const Text(
50                 "Drag Me",
51                 style: TextStyle(color: Colors.white),
52               ),
53             ),
54           ),
55         ],
56       ),
57     );
58   }
59 }
60 }
```

OUTPUT:



20260805-0418-17.4941690.mp4

5) DRAG IN ANY DIRECTION:

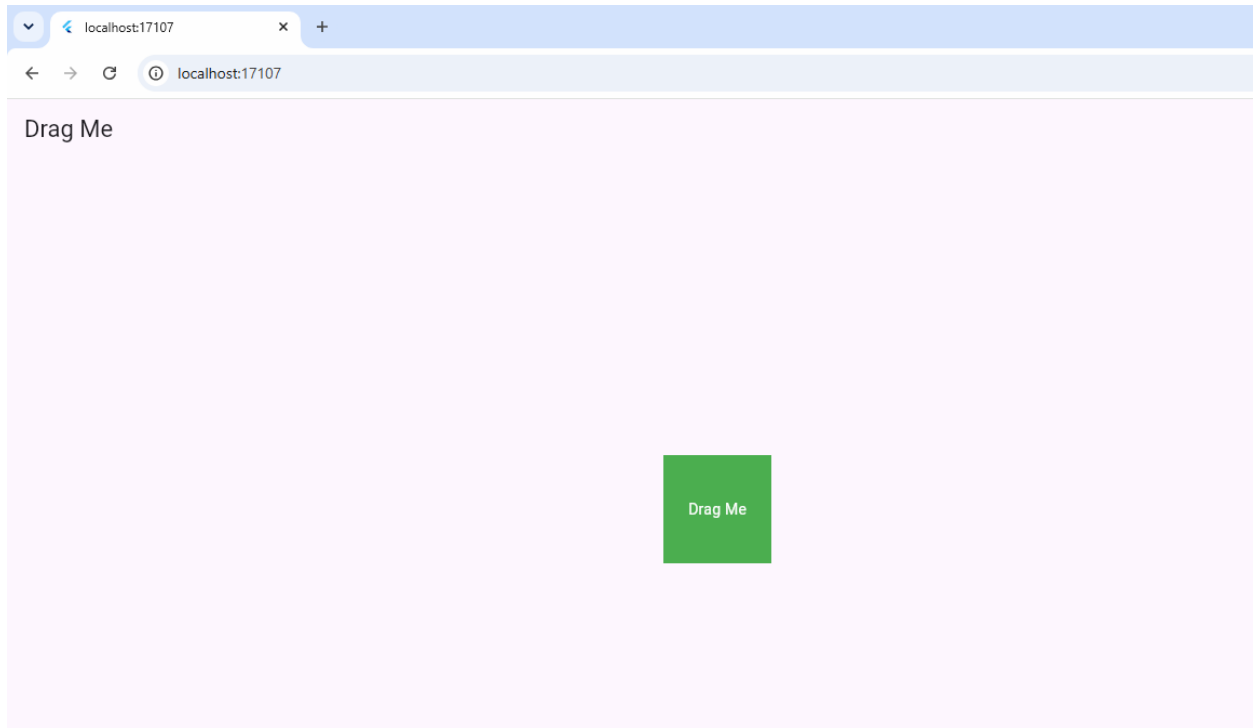
CODE:

```
Double Tap & Long Press.dart  Horizontal Drag Only.dart  Drag in Any Direction.dart x  Pinch to Zoom.dart  EXT...
```

```
1  import 'package:flutter/material.dart';
2
3  void main() {
4    runApp(const MyApp());
5  }
6
7  class MyApp extends StatelessWidget {
8    const MyApp({super.key});
9
10   @override
11   Widget build(BuildContext context) {
12     return const MaterialApp(
13       debugShowCheckedModeBanner: false,
14       home: DragExample(),
15     );
16   }
17 }
18
19 class DragExample extends StatefulWidget {
20   const DragExample({super.key});
21
22   @override
23   State<DragExample> createState() => _DragExampleState();
24 }
25
26 class _DragExampleState extends State<DragExample> {
27   double x = 100;
28   double y = 100;
```

```
30   @override
31   Widget build(BuildContext context) {
32     return Scaffold(
33       appBar: AppBar(title: const Text("Drag Me")),
34       body: GestureDetector(
35         onPanUpdate: (details) {
36           setState(() {
37             x += details.delta.dx;
38             y += details.delta.dy;
39           });
40         },
41         child: Stack(
42           children: [
43             Positioned(
44               left: x,
45               top: y,
46               child: Container(
47                 width: 100,
48                 height: 100,
49                 color: Colors.green,
50                 alignment: Alignment.center,
51                 child: const Text(
52                   "Drag Me",
53                   style: TextStyle(color: Colors.white),
54                 ),
55               ),
56             ),
57           ],
58         ),
59       ),
60     );
61   }
62 }
```

OUTPUT:



20260805-0421-17.8098835.mp4

6) VERTICAL DRAG ONLY:

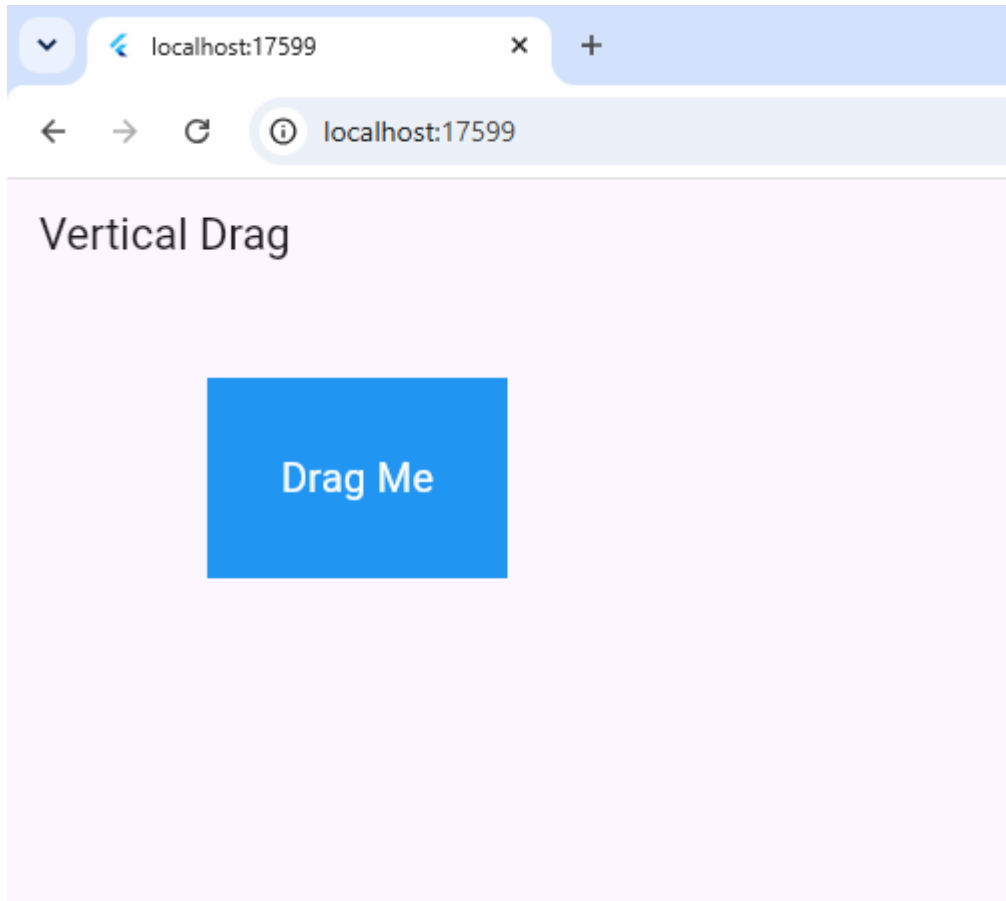
CODE:

```
Horizontal Drag Only.dart Drag in Any Direction.dart Pinch to Zoom.dart EXTERNAL.dart

1  import 'package:flutter/material.dart';
2
3  void main() {
4    runApp(const MyApp());
5  }
6
7  class MyApp extends StatelessWidget {
8    const MyApp({super.key});
9
10   @override
11   Widget build(BuildContext context) {
12     return const MaterialApp(
13       debugShowCheckedModeBanner: false,
14       home: VerticalDragDemo(),
15     );
16   }
17 }
18
19 class VerticalDragDemo extends StatefulWidget {
20   const VerticalDragDemo({super.key});
21
22   @override
23   State<VerticalDragDemo> createState() => _VerticalDragDemoState();
24 }
25
26 class _VerticalDragDemoState extends State<VerticalDragDemo> {
27   double top = 200;
28 }
```

```
Horizontal Drag Only.dart Drag in Any Direction.dart Pinch to Zoom.dart EXT
28
29 @override
30 Widget build(BuildContext context) {
31   return Scaffold(
32     appBar: AppBar(
33       title: const Text("Vertical Drag"),
34     ),
35     body: Stack(
36       children: [
37         Positioned(
38           left: 100,
39           top: top,
40           child: GestureDetector(
41             onVerticalDragUpdate: (details) {
42               setState(() {
43                 top += details.delta.dy;
44               });
45             },
46             child: Container(
47               width: 150,
48               height: 100,
49               color: Colors.blue,
50               alignment: Alignment.center,
51               child: const Text(
52                 "Drag Me",
53                 style: TextStyle(
54                   color: Colors.white,
55                   fontSize: 20,
56                 ),
57               ),
58             ),
59           ),
60         ),
61       ],
62     ),
63   );
64 }
65 }
```

OUTPUT:



20260805-0424-04.9927868.mp4